Potential Critical Path Selection Based on a Time-Varying Statistical Timing Analysis Framework

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***Abstract*— Negative bias temperature instability along with the presence of process variations has resulted in time-varying path criticalities. To ensure reliable circuit operation, aging sensors are used at the end of potential critical paths (PCPs) for delay monitoring. Optimization of the number of delay sensors requires accurate computational models for prediction of criticality and selection of PCPs. We identify a path as a PCP if its maximum global criticality over the lifetime exceeds a certain threshold. However, the global criticality of a path could vary nonmonotonically over the lifetime of the device. In this paper, we propose a framework for time-varying statistical static timing analysis (TV-SSTA), wherein the circuit delay is obtained as a collection of time-varying canonicals with breakpoints in time which define the end of validity of one and the start of the next canonical. We show that the global criticality of any path will be maximum either at $t = 0$ or at these breakpoints. Hence, criticality computation and PCP selection need to be done only at these time points, which typically is less than four. The TV-SSTA is integrated with a previously proposed criticality computation technique to identify the PCPs and the results are validated against Monte Carlo simulations.**

***Index Terms*— Aging, negative bias temperature instability (NBTI), path criticality, potential critical path (PCP), statistical timing.**

I. Introduction

With technology scaling, circuit aging has become a major reliability concern in digital integrated circuits. Aging effects cause transistor parameters to degrade over time, significantly affecting the circuit performance. One of the dominant aging effects is negative bias temperature instability (NBTI). NBTI degrades threshold voltage of pMOS transistors in the inversion region, leading to a progressive increase in circuit delay. Consequently, a circuit that meets timing at the testing phase may fail beyond a certain operation time, thus limiting the lifetime of the device. In the presence of process variations, the gate delay distributions change over time depending on the stress factors. This can result in different paths becoming critical not just in different dies, but also at different time points, leading to a time-varying path criticality. Any path that can become critical during the expected lifetime of the circuit is referred to as a "potential critical path" (PCP).

The traditional method to deal with circuit aging is guardbanding. However, this is highly pessimistic as it assumes worst case stress conditions for all the gates and results in significant performance loss. Recently, aging sensors have been proposed for on-line circuit delay prediction in the postsilicon phase [1], [2]. The aging sensors are typically placed at the end of PCPs for delay tracking. Adaptive techniques such as frequency scaling, voltage scaling, and adaptive body bias are then employed to mitigate the effects of aging [3]. Owing to the overhead associated with the insertion of aging sensors, it is infeasible to track the delay of a large number of paths. Also, missing a PCP could result in circuit failure. Therefore, it is essential to have a path selection technique that identifies the optimal set of PCPs necessary for circuit delay tracking.

Several methods have been proposed for the selection of PCPs under NBTI for delay monitoring. The first category of techniques follows a two-step procedure. The first step is the generation of an initial path set. The PCPs are then selected from this initial set using certain path selection criteria. In [4]–[6], both these steps are performed based on static timing analysis (STA). The technique proposed in [4] performs a traditional STA at $t = 0$ and enumerates all paths with worst case delay (assuming 20% degradation) greater than the delay of the longest critical path (LCP). A path from this initial set is identified as critical if its aged delay, obtained using an NBTI aware timing library, exceeds the fresh delay of LCP with a timing margin. The method proposed in [5] is similar to that of [4], with the difference that the path delays are compared with respect to a clock period rather than a delay of LCP obtained from STA. The approach proposed in [6] is based on timing graph reduction and does not have a separate path filtering step. An edge in the graph is removed if all the paths passing through it is noncritical even under worst case aging conditions. All the paths passing through the remaining edges are then chosen for testing under aging.

The techniques proposed in [7]–[9] use the results from STA for the generation of initial path set. However, the selection of PCPs is done incorporating the effect of process variations. In [7], the mean and standard deviation of fresh (without aging) and aged delays are obtained for each path in the initial set. A path is chosen as a PCP if the probability that its aged delay distribution is greater than the fresh delay distribution of LCP from STA exceeds a certain threshold.

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Gomez and Champac [8] and [9] claim that the local criticality of any path will be maximum at the endpoints, at $t = 0$, or at the end of lifetime. For every pair of paths p_i and p_j in the initial set, the local criticality of p_i with respect to p_j is computed at the endpoints and p_i is removed if the maximum criticality is below the threshold. However, this is obvious only if the variation in standard deviation with time is negligible.

Another PCP selection technique incorporating process variations and based on graph reduction was proposed in [10]. Initially, all edges with a low probability of being a part of any PCP are removed. The remaining paths in the reduced timing graph are then enumerated. A path in this set is selected as a PCP if the probability that its maximal aged delay distribution is greater than the minimal aged delay distribution of the longest path from STA exceeds a given threshold. However, identification of the edges to be pruned is not simple if there are a large number of uncorrelated paths with similar delays.

All these techniques require an initial path set and it is not clear how many paths should be selected. To ensure that no critical paths are missed, large number of paths should be enumerated which is time-consuming. Moreover, PCP selection using STA will result in a large number of highly correlated paths so that monitoring any one of these paths is sufficient for delay tracking. Even though the path selection using local criticality takes into account the correlations, the local criticality is a good approximation only if the number of paths in the initial set is large. The accurate measure for path criticality is its global criticality [11]. Therefore, an alternative technique for PCP selection is to perform block-based statistical static timing analysis (SSTA) at several time instants over the lifetime of the circuit using aged device models [12]–[14]. For path selection, this is followed by criticality computation at these time points. However, the global criticality of paths could vary nonmonotonically over the lifetime of the device. This makes it difficult to figure out the time instants at which path selection and criticality computations should be performed. Moreover, existing statistical aged gate delay models have some limitations which are explained in more detail in Section II. In this paper, we address these issues. The main contributions of our paper can be summarized as follows.

1) We derive a simple time-varying canonical delay model that considers the correlation between process variations and NBTI. It also includes the intrinsic charge fluctuations, which are independent of process variations.
2) We develop a time-varying SSTA (TV-SSTA) framework to evaluate circuit delay as a collection of time-varying canonicals, with breakpoints in time. A breakpoint defines the time of switching from one canonical model to the other.
3) We show that the global criticality of any path will be maximum either at $t = 0$ or at the breakpoints in the circuit delay distribution. Therefore, criticality computation needs to be done only at these time points.
4) We make use of the hierarchical partitioning technique proposed in [15] at these time points to identify the PCPs with global criticality above a specified threshold.

The rest of this paper is organized as follows. In Section II, we develop a time-varying canonical model for the representation of gate delays under NBTI. Section III discusses the proposed technique for TV-SSTA and the implementation details. In Section IV, we derive the properties of time-varying path criticality. The proposed algorithm for PCP selection is discussed in Section V. Section VI contains a description of the overall framework. Section VII presents the results and Section VIII concludes this paper.

II. Time-Varying Canonical Delay Model

A. Threshold Voltage Degradation

The threshold voltage degradation due to NBTI at high frequencies is modeled as [16]

$$V_{\text{th,nbti}}(t) = A_{\text{nbti}}\ (V_{\text{th}}(t=0))\ \alpha_s^n t^n \tag{1}$$

where α_s equals $\alpha/(1-\alpha)$, α is the stress factor indicating the fraction of time the transistor is under stress and $n = 1/6$. The coefficient A_{nbti} depends on the operating conditions and initial threshold voltage at time $t = 0$ where

$$V_{\text{th}}(t=0) = V_{\text{th0}} + V_{\text{th,PV}} \tag{2}$$

where V_{th0} is the nominal threshold voltage and $V_{\text{th,PV}} = \mathcal{N}(0, \sigma^2_{V_{\text{th,PV}}})$ is a zero-mean random variable representing the process variations in threshold voltage. The dependence of A_{nbti} on process variations is modeled as [17]

$$A_{\text{nbti}}(V_{\text{th}}(t=0)) = A_0(1 + S_A V_{\text{th,PV}}) \tag{3}$$

where $A_0 = A_{\text{nbti}}(V_{\text{th0}})$ and S_A indicates the sensitivity of A_{nbti} to V_{th} variations normalized with respect to A_0. Clearly, the two random variables, $V_{\text{th,nbti}}(t)$ and $V_{\text{th,PV}}$, are correlated.

In addition to this, there is an independent intrinsic component within NBTI due to the random charge fluctuations of the generated interface traps. The variance of the intrinsic component is modeled as [18]

$$\sigma^2_{\text{int}} = K\mu\{V_{\text{th,nbti}}(t)\} \tag{4}$$

where $\mu\{V_{\text{th,nbti}}(t)\}$ is the mean of threshold voltage degradation, $K = qt_{\text{ox}}/(\epsilon_{\text{ox}}A_G)$ and A_G is the transistor area.

The statistical aged gate delay models in [12], [17], and [19] considered the correlations between process variations and NBTI, but neglected the variations due to the intrinsic charge fluctuations. The models proposed in [13], [18], and [20] included the random charge fluctuations within NBTI. However, the threshold voltage degradation was modeled as a function of the nominal threshold voltage. The method proposed in [21] used Monte Carlo-based transistor-level simulations to study the effect of process variations on aging and, hence, is not suited for the analysis of large circuits. The approach in [14] considered run-time voltage and temperature variations and modeled BTI as a log-normal distribution. Although comprehensive, it is not in the standard SSTA format since the gate delay was modeled as a mix of normal and log normal distributions.

Considering this, we first derive a simple time-varying model for threshold voltage that takes into account the correlation with process variations as well as the intrinsic fluctuations within NBTI and can very easily be incorporated into the

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canonical delay model used in SSTA. Using iterated expectations and law of total variance, the mean and variance of threshold voltage degradation due to NBTI is computed as follows:

$$\begin{aligned}\mu\{V_{\text{th,nbti}}(t)\} &= \mathrm{E}_{V_{\text{th,PV}}}\{\mathrm{E}\{V_{\text{th,nbti}}(t)|V_{\text{th,PV}}\}\} \\ &= A_0\alpha_s^n t^n \qquad (5)\\ \sigma^2\{V_{\text{th,nbti}}(t)\} &= \mathrm{Var}_{V_{\text{th,PV}}}\{\mathrm{E}\{V_{\text{th,nbti}}(t)|V_{\text{th,PV}}\}\} \\ &\quad + \mathrm{E}_{V_{\text{th,PV}}}\{\mathrm{Var}\{V_{\text{th,nbti}}(t)|V_{\text{th,PV}}\}\} \\ &= \left(A_0 S_A \alpha_s^n t^n\right)^2 \sigma^2_{V_{\text{th,PV}}} + \mathrm{K}A_0\alpha_s^n t^n. \qquad (6)\end{aligned}$$

Let $\Delta V_{\text{th}}(t)$ denote the overall deviation in threshold voltage from the nominal value. Hence

$$\mu\{\Delta V_{\text{th}}(t)\} = \mathrm{E}\{V_{\text{th,PV}} + V_{\text{th,nbti}}(t)\} = A_0\alpha_s^n t^n. \qquad (7)$$

Since $V_{\text{th,PV}}$ and $V_{\text{th,nbti}}(t)$ are correlated, the variance of $\Delta V_{\text{th}}(t)$ can be derived as follows:

$$\begin{aligned}\sigma^2\{\Delta V_{\text{th}}(t)\} &= \sigma^2_{V_{\text{th,PV}}} + \sigma^2_{V_{\text{th,nbti}}(t)} + 2\mathrm{Cov}\{V_{\text{th,PV}} V_{\text{th,nbti}}\} \\ &= \left(1 + A_0 S_A \alpha_s^n t^n\right)^2 \sigma^2_{V_{\text{th,PV}}} + \mathrm{K}A_0\alpha_s^n t^n \qquad (8)\end{aligned}$$

where $\mathrm{Cov}\{V_{\text{th,PV}} V_{\text{th,nbti}}\}$ is the covariance between process variations and NBTI-induced threshold voltage degradation.

In the canonical delay model used in the statistical timing analysis, $V_{\text{th,PV}}$ is expressed in terms of principal components for modeling the spatial correlations. Therefore, we model $\Delta V_{\text{th}}(t)$ in terms of the principal components (X_i) as

$$\Delta V_{\text{th}}(t) = A_0\alpha_s^n t^n + \left(1 + A_0 S_A \alpha_s^n t^n\right) \sum_{i=1}^{N} S_{X,i} X_i + S_{\text{int}} t^{\frac{n}{2}} R_{\text{int}} \qquad (9)$$

where $S_{\text{int}} = (\mathrm{K}A_0\alpha_s^n)^{1/2}$ and R_{int} is an independent component that models the random charge fluctuations in NBTI.

As indicated by (1), the threshold voltage degradation due to NBTI is proportional to coefficient A_{nbti} which in turn depends on the operating conditions. Since a higher supply voltage and temperature result in higher rate of V_{th} degradation [22], a worst case operating condition of ($V_{\text{dd}} = 1.32V$, $T = 125°\text{C}$) and ($V_{\text{dd}} = 1.1V$, $T = 125°\text{C}$) for 90- and 45-nm technology, respectively, is considered for our analysis.

The threshold voltage degradation under NBTI also depends on the stress factor (α) and, hence, is different for different workload conditions at the primary inputs. In this paper, we ignore any variations in the input workload with time. As the input workload is unknown, a static probability (SP) of 0.5 is assumed at the primary inputs. Our simulation results show that the circuit delay degradation under various input SP conditions is typically within 5% of the SP = 0.5 assumption. The SPs, as well as the pairwise signal correlations at the gate inputs, are calculated from the value change dump (VCD) file after performing logic simulation. The stress factors are then computed using the technique proposed in [23].

B. Gate Delay Model

Ideally, for an n-input gate, the delay degradation of any input–output arc depends on the threshold voltage degradation of all the pMOS transistors. However, similar to the existing statistical timing analysis techniques under NBTI [14], [17], we assume that the delay degradation of arc (x_i, y) depends only on the threshold voltage degradation of transistor M_i connected to x_i. Using the first-order Taylor series expansion of alpha power law model, the dependence of gate delay on time-varying V_{th} variations is modeled as [17]

$$d(t) = \mu_{d0} + S_V \Delta V_{\text{th}}(t) \qquad (10)$$

where μ_{d0} is the nominal delay and S_V is the sensitivity of gate delay to changes in threshold voltage. Therefore, the time-varying canonical delay model can be obtained from (9) as

$$\begin{aligned}d(t) = \mu_{d0} + d_t t^n &+ \sum_{i=1}^{N} (S_{d_{X,i0}} + d_{X,i} t^n) X_i \\ &+ \sum_{j=1}^{M} (S_{d_{R,j0}} + d_{R,j} t^n) R_j + d_{R,\text{int}} t^{\frac{n}{2}} R_{\text{int}} \qquad (11)\end{aligned}$$

where μ_{d0} denotes the nominal gate delay at $t = 0$. $S_{d_{X,i0}}$ and $S_{d_{R,j0}}$ denote the sensitivities associated with process variations (X_i) and random components (R_j) at $t = 0$. d_t, $d_{X,i}$, and $d_{R,j}$ denote the sensitivities associated with time dependence which, in turn, depend on the parameters S_V, S_A, A_0, and α. $d_{R,\text{int}} t^{\frac{n}{2}}$ denotes the sensitivity associated with the intrinsic component R_{int}. In comparison with the extended canonical form defined in [24], there are two main differences, namely, (a) the mean and sensitivities are functions of time and are of the form $x_0 + x_t t^n$ and (b) there is an additional random component (R_{int}) representing the intrinsic fluctuations in NBTI. Note that only rise delays degrade due to NBTI and falling gate delays are independent of time.

III. Time-Varying SSTA

A block-based TV-SSTA is used to find the arrival time at each node as a function of time. Sections III-A and III-B describe SUM and MAX operations on delays expressed in the time-varying canonical format.

A. SUM Operation

For inputs A and B in the time-varying canonical form, we want to represent $C = \mathrm{SUM}(A, B)$ also in the same format. Since SUM is a linear operator, the mean and principal and random component sensitivities at the output can be obtained by adding the corresponding input sensitivities. The intrinsic component sensitivity is given by $c_{R,\text{int}} t^{(n/2)} = (a^2_{R,\text{int}} + b^2_{R,\text{int}})|^{1/2} t^{(n/2)}$.

B. MAX Operation

Given inputs A and B in time-varying canonical form, the aim is to find $C = \mathrm{MAX}(A, B)$ also in the time-varying canonical format. Clark [25] proposed an analytical model for the mean and variance (μ_c, σ_c^2) of maximum of two normally distributed random variables. In this section, we extend it to include the time-varying means and sensitivities.

For a timing quantity A expressed in the time-varying canonical model, the time variation of mean and sensitivities

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are as in (11). Using these, we obtain the time-varying variance as

$$\sigma_a^2(t) = \sum_{i=1}^{N}(S_{a_{X,i0}}+a_{X,i}t^n)^2+\sum_{j=1}^{M}(S_{a_{R,j0}}+a_{R,j}t^n)^2+a_{R,\text{int}}^2t^n = \sigma_{a0}^2 + \Delta\sigma_a^2(t) \quad (12)$$

where σ_{a0}^2 represents the variance of A at $t = 0$. $\Delta\sigma_a^2(t)$ denotes the deviation in variance of A from the initial value at $t = 0$ and consists of first-order and second-order terms. From (9)–(11), the ratio of magnitude of second-order to first-order term is of the order of $10^{-3}t^n$, which is negligible over the lifetime of the device. Therefore, we approximate $\Delta\sigma_a^2(t)$ as

$$\Delta\sigma_a^2(t) = \sigma_{a_t}^2 t^n \quad \text{where}$$
$$\sigma_{a_t}^2 = \sum_{i=1}^{N}2S_{a_{X,i0}}a_{X,i} + \sum_{j=1}^{M}2S_{a_{R,j0}}a_{R,j} + a_{R,\text{int}}^2. \quad (13)$$

The tightness probability between two timing quantities A and B is defined as $\Phi(\beta)$, which is the cumulative density function of standard normal distribution [25]. With aging, β is a function of time given by

$$\beta(t) = \frac{\mu_a(t) - \mu_b(t)}{\theta(t)} \quad (14)$$

where

$$\theta(t) = \sqrt{\sigma_a^2(t) + \sigma_b^2(t) - 2\rho(t)\sigma_a(t)\sigma_b(t)}. \quad (15)$$

The correlation coefficient $\rho(t)$ captures the topological and spatial correlations between A and B at time instant t. We ignore the marginal changes in correlations between the gates due to differential stress and assume ρ to be independent of time. However, the mean and variance of A and B are functions of time thereby making β and $\Phi(\beta)$ also time-varying.

The mean and sensitivities at the output of MAX operation depend on tightness probability, which is strongly nonlinear as it approaches 0 or 1. Therefore, we divide the tightness probability into four regions as shown in Fig. 1. Within each region, we can find a time-varying canonical representation for the output. In region (1), $\Phi(\beta) \approx 0$, and hence, MAX(A, B) can be approximated as B. Similarly, in region (4), $\Phi(\beta) \approx 1$ and MAX(A, B) approximately equals A. In these two cases, the result of the MAX operation is in the canonical form.

If β lies within region (2) or (3), the output canonical is evaluated using a first-order Taylor series expansion. This is a good approximation since the variation in mean and standard deviation over the lifetime is typically small, less than 15% and 5%, respectively. Assume β varies from β_i to β_j within a region in a time interval $[t_i, t_j]$. The evaluation of time-varying mean and sensitivities of the output canonical is discussed in Sections III-B1–III-B3.

1) Mean as a Function of Time ($\mu_c(t) = \mu_c(t_i) + \Delta\mu_c(t)$): By Clark's approximation [25], the mean delay at the output of MAX operation at $t = t_i$ is given by

$$\mu_c(t_i) = \mu_a(t_i)\Phi(\beta_i) + \mu_b(t_i)\Phi(-\beta_i) + \theta_i\phi(\beta_i) \quad (16)$$

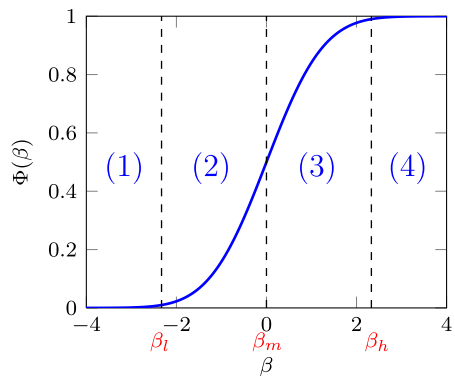

Fig. 1. Four regions in tightness probability domain determined by β_l, β_m, and β_h.

where $\theta_i = \theta(t_i)$. The deviation in mean from the initial value at t_i is $\Delta\mu_c(t) = c_t(t^n - t_i^n)$. Assuming small deviations, c_t is computed using the first-order Taylor series expansion of $\mu_c(t_i)$ with respect to the means and variances and is given by

$$c_t = \Phi(\beta_i)a_t + \Phi(-\beta_i)b_t + \frac{\phi(\beta_i)}{2\theta_i}\left(\left(1-\rho\frac{\sigma_{bi}}{\sigma_{ai}}\right)\sigma_{a_t}^2 + \left(1-\rho\frac{\sigma_{\text{ai}}}{\sigma_{\text{bi}}}\right)\sigma_{b_t}^2\right) \quad (17)$$

where σ_{ai} and σ_{bi} denote the standard deviations of A and B at $t = t_i$, respectively.

2) Principal Component Sensitivity as Function of Time ($S_{c_{X,i}}(t) = S_{c_{X,i}}(t_i) + \Delta S_{c_{X,i}}(t)$): The sensitivity to principal component at $t = t_i$ is given by [26]

$$S_{c_{X,i}}(t_i) = S_{a_{X,i}}(t_i)\Phi(\beta_i) + S_{b_{X,i}}(t_i)\Phi(-\beta_i). \quad (18)$$

As before, the deviation in principal component sensitivity from the initial value at $t = t_i(\Delta S_{c_{X,i}}(t) = c_{X,i}(t^n - t_i^n))$ is computed using the first-order Taylor series expansion. The variation of $\beta(t)$ within the time interval is approximated as

$$\beta(t) \approx \beta_i + \beta_t(t^n - t_i^n) \quad (19)$$

where

$$\beta_t = \frac{a_t - b_t}{\theta_i} + \frac{(\mu_b(t_i) - \mu_a(t_i))}{2\theta_i^3} \times \left(\left(1-\rho\frac{\sigma_{\text{bi}}}{\sigma_{\text{ai}}}\right)\sigma_{a_t}^2 + \left(1-\rho\frac{\sigma_{\text{ai}}}{\sigma_{bi}}\right)\sigma_{b_t}^2\right). \quad (20)$$

As a result, $c_{X,i}$ is given by

$$c_{X,i} = \Phi(\beta_i)a_{X,i} + \Phi(-\beta_i)b_{X,i} + (S_{a_{X,i}}(t_i) - S_{b_{X,i}}(t_i))\phi(\beta_i)\beta_t. \quad (21)$$

The sensitivity coefficient associated with random component is derived similarly.

3) Sensitivity of Intrinsic Component $c_{R,int}t^{(n/2)}$: Given the sensitivities of A and B, the sensitivity associated with the intrinsic component in C is calculated as follows:

$$c_{R,\text{int}}t^{\frac{n}{2}} = t^{\frac{n}{2}}\sqrt{[a_{R,\text{int}}\Phi(\beta(t))]^2 + [b_{R,\text{int}}\Phi(-\beta(t))]^2}. \quad (22)$$

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Since we wish to express the output of MAX operation in time-varying canonical form, we approximate $c_{R,\text{int}}$ as

$$c_{R,\text{int}} \approx \sqrt{(a_{R,\text{int}}\Phi(\beta_i))^2 + (b_{R,\text{int}}\Phi(-\beta_i))^2}. \tag{23}$$

Therefore, each time-varying quantity is represented in the form $x(t) = x(t_i) + x_t(t^n - t_i^n)$ which is in accordance with the canonical model defined in (11). If β increases/decreases over time and crosses over to the next region, then there is a switch from one canonical model to the other and the corresponding breakpoint in time is evaluated using (19). Thus, the output of the MAX operator is stored as a collection of time-varying canonicals with different canonicals being valid over different time intervals.

C. Implementation

Let the inputs to MAX operation be A_c and B_c, where A_c and B_c are a collection of time-varying canonicals with arbitrary number of break points and defined over the lifetime (t_{life}). Let $t_{A1}, t_{A2}, \ldots t_{\text{life}}$ be the break points in A_c and $t_{B1}, t_{B2}, \ldots t_{\text{life}}$ be the break points in B_c. In the first step, the break points in A_c and B_c together are sorted in the increasing order to find the break points in the output of MAX operation. The input canonicals associated with each of these time intervals are also identified. For a given time interval $[t_i, t_j]$ with input canonicals A_i and B_i, the identification of break points within this interval and the evaluation of MAX is outlined in Algorithm 1.

We start with defining β_l, β_m, and β_h, the points which determine the four regions in $\Phi(\beta)$ curve. β_m is fixed at zero and β_l and β_h correspond to the points where $\Phi(\beta)$ is 0.01 and 0.99, respectively (step 1). The next step is to identify if there is any switching of regions during the time interval $[t_i, t_j]$ over which the inputs are defined. The values of β_l, β_m, and β_h are compared against β_i and β_j to check if there is any switching of regions. If so, the time at which switching of regions occur is evaluated and this is added to the list of breakpoints. The list of breakpoints is then sorted in ascending order and time t_j is added as the last breakpoint (steps 2–13).

Once the breakpoints are obtained, we evaluate the output canonical in each of the identified time intervals (steps 14–30). This is done as discussed in Section III-B. If $\beta \leq \beta_l$ in the time interval under consideration, the output canonical is the same as input B_i. Similarly, if $\beta \geq \beta_h$ in the time interval, the output canonical equals A_i. If $\beta_l < \beta \leq \beta_m$ or $\beta_m < \beta < \beta_h$ in the time interval, the output canonical is computed using the formulas derived in Sections III-B1–III-B3.

To keep a limit on the number of breakpoints, a check is performed after every MAX operation to see if any of the neighboring canonicals can be merged without introducing significant error. Let X and Y be two neighboring canonicals with time intervals $[t_1, t_2]$ and $[t_2, t_3]$, respectively. The two canonicals are merged to obtain canonical X valid over the time interval $[t_1, t_3]$ if the error in mean and standard deviation at t_3 introduced in this approximation is within 1%.

Algorithm 1 Maximum of Two Time-Varying Canonicals

Input : A_i and B_i
Output: $C_i = MAX(A_i, B_i)$
1: $\beta_l \leftarrow -2.33$, $\beta_m \leftarrow 0$ and $\beta_h \leftarrow 2.33$
2: $\beta_i \leftarrow \beta(t = t_i)$
3: Find β_t using (20)
4: $\beta_j \leftarrow \beta_i + \beta_t(t_j^n - t_i^n)$
5: Initialise $\beta_{brk} = [\beta_l, \beta_m, \beta_h]$; t_{brk} = [ ]
6: **for** $\beta_b \in \beta_{brk}$ **do**
7: **if** $\beta_i < \beta_b < \beta_j$ **then**
8: Solve $\beta_b = \beta_i + \beta_t(t_b^n - t_i^n)$ to find t_b
9: Add t_b to t_{brk}
10: **end if**
11: **end for**
12: Sort t_{brk} in ascending order
13: Add t_j to t_{brk}
14: $k \leftarrow 0$
15: Initialise C_i =[ ]
16: **while** $k < len(t_{brk})$ **do**
17: **if** $k == 0$ **then**
18: Time interval, TI = $[t_i, t_{brk}[k]]$
19: **else**
20: Time interval, TI = $[t_{brk}[k-1], t_{brk}[k]]$
21: **end if**
22: **if** $\beta \leq \beta_l$ in TI **then**
23: Add B_i to C_i
24: **else if** $\beta \geq \beta_h$ in TI **then**
25: Add A_i to C_i
26: **else**
27: Evaluate output canonical using formulas derived in Sections (III-B1) to (III-B3) and add to C_i
28: **end if**
29: $k += 1$
30: **end while**
31: return C_i

IV. Time-Varying Path Criticality

The global criticality (Q_p) of path p is defined as the probability that delay of the path (D_p) exceeds the complementary path delay (CD_p) [15]. With aging, all the gate delays vary with time, thereby making the path delay and complementary path delay also time-varying. Thus the global criticality of a path under aging is a function of time that is given by

$$Q_p(t) = P(D_p(t) > \text{CD}_p(t)) \tag{24}$$

where $D_p(t)$ and $\text{CD}_p(t)$ are Gaussian random variables with time-varying means and variances. A path p in the circuit is treated as a PCP if its maximum global criticality over the lifetime exceeds a threshold, that is

$$\max_{0 \leq t \leq t_{\text{life}}} \{Q_p(t)\} \geq \gamma \tag{25}$$

where t_{life} is the expected lifetime and γ is the criticality threshold. This requires the evaluation of maximum global criticality of a path over the lifetime of the circuit. However, the global path criticality can vary nonmonotonically over time as indicated in Fig. 2. These criticality values are obtained using Monte Carlo simulations from 0 to 10 years in steps of 1 year.

It is infeasible to compute the criticality at each time instant within the lifetime for determining whether the path is potentially critical or not. We derive the following properties of time-varying path criticalities under aging to overcome this problem.

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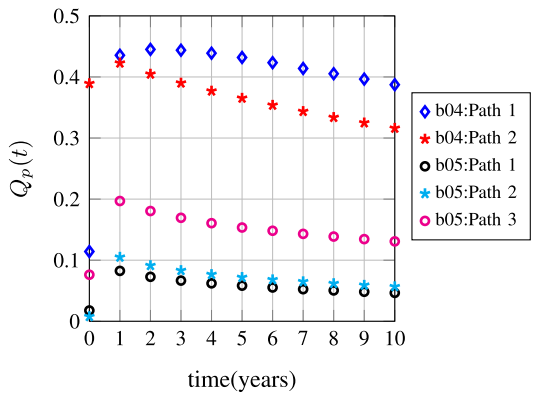

Fig. 2. Nonmonotonic time-varying path criticalities of some paths in b04 and b05 circuits obtained using Monte Carlo simulations in 90-nm technology.

Property 1: If $D_p(t) = \mathcal{N}(\mu_{D_p}(t), \sigma^2_{D_p}(t))$ and $\mathrm{CD}_p(t) = \mathcal{N}(\mu_{\mathrm{CD}_p}(t), \sigma^2_{\mathrm{CD}_p}(t))$ denote the time-varying path delay and complementary path delay, respectively, then the time-varying global path criticality is given by $\Phi(\beta(t))$ where

$$\beta(t) = \frac{\mu_{D_p}(t) - \mu_{\mathrm{CD}_p}(t)}{\sqrt{\sigma^2_{D_p}(t) + \sigma^2_{\mathrm{CD}_p}(t) - 2\rho(t)\sigma_{D_p}(t)\sigma_{\mathrm{CD}_p}(t)}} \quad (26)$$

and $\rho(t)$ is the correlation coefficient between path delay and complementary delay at time t.

Proof: The time-varying global path criticality is given by

$$Q_p(t) = P(D_p(t) > \mathrm{CD}_p(t)). \quad (27)$$

Based on the delay model given by (11), $D_p(t)$ and $\mathrm{CD}_p(t)$ are jointly Gaussian random variables with time-varying means, variances, and correlation coefficient. Let $D(t)$ denote the delay difference random variable $D_p(t) - \mathrm{CD}_p(t)$. Therefore, at any given time t

$$Q_p(t) = P(D(t) > 0) = \Phi\left(\frac{\mu_d(t)}{\sigma_d(t)}\right) = \Phi(\beta(t)) \quad (28)$$

where

$$\mu_d(t) = \mu_{D_p}(t) - \mu_{CD_p}(t) \quad (29)$$

and

$$\sigma_d(t) = \sqrt{\sigma^2_{D_p}(t) + \sigma^2_{\mathrm{CD}_p}(t) - 2\rho(t)\sigma_{D_p}(t)\sigma_{\mathrm{CD}_p}(t)}. \quad (30)$$

□

Property 2: If $A_i(t)$ and $B_i(t)$ are two time-varying canonicals valid over the time interval $[t_i, t_j]$ with β as in (19), then criticality of A_i with respect to B_i will be maximum either at t_i or t_j.

Proof: Let $Q_{A_iB_i}(t)$ denote the criticality of A_i with respect to B_i at any time instant $t \in [t_i, t_j]$. We have to show that

$$\max_{t_i \le t \le t_j} \{Q_{A_iB_i}(t)\} = \max\{Q_{A_iB_i}(t_i), Q_{A_iB_i}(t_j)\}. \quad (31)$$

Given that $A_i(t)$ and $B_i(t)$ are represented in the time-varying canonical form with $\beta(t)$ as in (19)

$$\text{If } \beta_t \ge 0, \quad \max_{t_i \le t \le t_j} \{\beta(t)\} = \beta(t_j)$$
$$\text{If } \beta_t < 0, \quad \max_{t_i \le t \le t_j} \{\beta(t)\} = \beta(t_i)$$
$$\begin{aligned}\max_{t_i \le t \le t_j} \{Q_{A_iB_i}(t)\} &= \max_{t_i \le t \le t_j} \{\Phi(\beta(t))\} \\ &= \max\{\Phi(\beta(t_i)), \quad \Phi(\beta(t_j))\} \\ &= \max\{Q_{A_iB_i}(t_i), \quad Q_{A_iB_i}(t_j)\}.\end{aligned}$$

□

Property 3: If $A(t)$ and $B(t)$ are a collection of time-varying canonicals over the lifetime, the break points in $\max\{A(t), B(t)\}$ will be a super set of the union of break points in $A(t)$ and $B(t)$.

Proof: Let $t_b(A)$ be the break points in $A(t)$ and $t_b(B)$ be the break points in $B(t)$. Let $t_{A\cup B} = t_b(A) \cup t_b(B)$ denote the union of break points in $A(t)$ and $B(t)$. If $C(t) = \max(A(t), B(t))$, we want to show that $t_b(C) \supseteq t_{A\cup B}$.

According to the implementation details in Section III-C, MAX operation only introduces additional break points. Therefore, break points in $C(t)$ will always be a superset of the break points in $A(t)$ and $B(t)$, i.e., $t_b(C) \supseteq t_{A\cup B}$. □

Property 4: If $A(t)$ and $B(t)$ are a collection of time-varying canonicals over the lifetime, then the criticality of A with respect to B will be maximum either at $t = 0$ or at the break points of $\max\{A(t), B(t)\}$.

Proof: Let $Q_{AB}(t)$ denote the criticality of A with respect to B at any time instant $t \in [0, t_{\mathrm{life}}]$. We have to show that

$$\max_{0 \le t \le t_{\mathrm{life}}} \{Q_{AB}(t)\} = \max_{t_i \in t_{\mathrm{bmax}}} \{Q_{AB}(0), Q_{AB}(t_i)\} \quad (32)$$

where t_{bmax} denote the break points in $\max\{A(t), B(t)\}$. This property follows directly from properties 2 and 3. □

Although this limits the points at which the criticality needs to be computed, it can be still very large as the set of breakpoints for each path delay and its corresponding complementary path delay could be different. The following property reduces this set significantly.

Property 5: The time-varying global criticality of any path will be maximum either at $t = 0$ or at any of the break points in circuit delay distribution.

Proof: Consider a path i in the circuit and let $D_{p,i}(t)$ and $\mathrm{CD}_{p,i}(t)$ denote its delay and complementary path delay, respectively. If there are N paths in the circuit, complementary path delay is given by

$$\mathrm{CD}_{p,i}(t) = \max_{\substack{1 \le j \le N \\ j \ne i}} \{D_{p,j}(t)\}. \quad (33)$$

The circuit delay distribution (D_{ckt}) is the statistical maximum of the delay of all the paths in the circuit. Therefore,

$$D_{\mathrm{ckt}}(t) = \max\{D_{p,i}(t), \mathrm{CD}_{p,i}(t)\}. \quad (34)$$

Making use of (34) and Property 4, we get

$$\max_{0 \le t \le t_{\mathrm{life}}} \{Q_{p,i}(t)\} = \max_{t_i \in t_b(D_{\mathrm{ckt}})} \{Q_{p,i}(0), Q_{p,i}(t_i)\} \quad (35)$$

where $t_b(D_{\mathrm{ckt}})$ denotes the break points in the circuit delay distribution. This holds true for any path i in the circuit.

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Therefore, the global criticality of any path in the circuit will be maximum either at $t = 0$ or at the break points in the circuit delay distribution. □

As a result of this property, we need to now evaluate criticalities and identify PCPs only at $t = 0$ and at the breakpoints of circuit delay distribution. This typically is less than four as will be seen in Section VII.

V. Statistical PCP Selection

The objective of the work is to identify the set of PCPs (S_{PCP}) with maximum global criticality over the lifetime greater than a threshold (γ). Mathematically

$$\text{A path } p \in S_{\text{PCP}} \text{ if } \max_{0 \le t \le t_{\text{life}}} \{Q_p(t)\} \ge \gamma. \tag{36}$$

By Property 5 given in Section IV, the global path criticality will be maximum either at $t = 0$ or at the break points in the circuit delay distribution. Therefore, we can conclude that a search for critical paths at these time points is sufficient to identify the PCPs under aging.

We make use of the hierarchical partitioning algorithm proposed in [15] to identify the paths with global criticality above a specified threshold at a given time instant. This algorithm integrates path selection and global criticality computation. This is done hierarchically with the circuit being split into disjoint groups at each level. The extension of the hierarchical partitioning algorithm for identification of PCPs under NBTI is outlined in Algorithm 2.

The inputs to the PCP selection algorithm are the circuit graph (G) and the desired criticality threshold (γ). In order to take into account the errors in the TV-SSTA formulation and path criticality computation, we choose the threshold for hierarchical partitioning to be lower than the desired threshold depending on the average error value (Δ). We also go through two iterations. In the first iteration, we identify the paths with criticality above the threshold at $t = 0$ and at the breakpoints in circuit delay distribution using the hierarchical partitioning algorithm. Once this is done, we compare the sum of path criticalities at each of the breakpoints against the value at $t = 0$. If the value has dropped significantly, it is possible that we have missed some critical paths at that breakpoint. This could happen due to the errors inherent in the SSTA formulation. Therefore, we reduce the threshold further and go through a second iteration to identify an additional set of critical paths.

A detailed description of the algorithm is given in the following. We start with a time-varying forward SSTA to obtain the circuit delay distribution as a function of time. The breakpoints in the circuit delay along with $t = 0$ form the time instants (T_{brk}) at which hierarchical partitioning should be done for the PCP selection (steps 1 and 2). In the next step, we perform the first level of partitioning based on output nets (steps 3–5). All the paths passing through a given output net i form a level (1) group and the arrival time at the output net ($AT_i(t)$) indicates its time-varying delay distribution.

A level (1) group G_k needs to be partitioned further at any time instant t_i only if its global criticality $Q_{g,k}(t_i)$ exceeds the threshold γ. This is because the criticality of any path within the group cannot be greater than the group criticality. For each time instant t_i, pruning based on local criticality is performed to remove the nondominant groups whose criticality is guaranteed to be lower than the threshold (step 8). Clustering of similar groups is also done to merge groups that have high topological correlation as discussed in [15]. The criticality of the dominant level (1) groups are then evaluated (steps 12 and 13). If the criticality of group G_k exceeds the threshold at t_i, then group G_k should be partitioned at t_i. This is done in two iterations as explained earlier. In the first iteration (steps 15–21), the threshold is set at a value Δ below the desired value and the set of critical paths at t_i are obtained. The threshold Δ is chosen based on the average error in path criticality computation. If the sum of path criticalities has dropped by Δ in comparison to the value at $t = 0$, a second round of hierarchical partitioning is done after lowering the threshold further (steps 23–32). Finally, the set of PCPs is the

Algorithm 2 PCP Selection Using Hierarchical Partitioning

Input : $G(V, E)$: Graph representing circuit netlist
γ : Criticality threshold
Output: S_{PCP} : Set of potential critical paths

/* Time-varying SSTA */
1: Perform time-varying SSTA to get circuit delay $D_{ckt}(t)$
2: $T_{brk} \leftarrow \{0\} \cup t_b(D_{ckt})$
/* Level (1) partitioning based on output nets */
3: **for** each net $i \in$ output nets N_{out} **do**
4: $D_i(t) \leftarrow AT_i(t)$
5: **end for**
/* Hierarchical partitioning of critical level (1) groups */
6: **for** each time $t_i \in T_{brk}$ **do**
7: Find gate delays and arrival times by substituting $t = t_i$ in the time-varying canonical form
8: $G_{dom}(t_i) \leftarrow$ Dominant groups at t_i after pruning and clustering of similar groups
9: $Qlist \leftarrow \{\}$ //Initialize list of group criticalities at t_i
10: $\Sigma Q_P(t_i) \leftarrow 0$ //Initialize sum of path criticalities at t_i
11: **for** each group $G_k \in G_{dom}(t_i)$ **do**
12: Evaluate complementary delay $CD_{g,k}(t_i)$
13: $Q_{g,k}(t_i) = P(D_{g,k}(t_i) > CD_{g,k}(t_i))$
14: Add $Q_{g,k}(t_i)$ to $Qlist$
/* Iteration : 1 */
15: $\gamma_1 = \gamma - \Delta$
16: **if** $Q_{g,k}(t_i) \ge \gamma_1$ **then**
17: $S_P(t_i) \leftarrow$ Hier_partitioning(G_k,t_i,$Q_{g,k}(t_i)$,γ_1)
18: $S_{PCP} \leftarrow S_{PCP} \cup S_P(t_i)$
19: $Q_P(t_i) \leftarrow$ Sum of criticalities of paths in $S_P(t_i)$
20: $\Sigma Q_P(t_i) += Q_P(t_i)$
21: **end if**
22: **end for**
/* Iteration : 2 */
23: **if** $t_i \neq 0$ and $\Sigma Q_P(0) - \Sigma Q_P(t_i) > \Delta$ **then**
24: Lower the threshold to $\gamma_2 = \gamma - 2\Delta$
25: **for** each group $G_k \in G_{dom}(t_i)$ **do**
26: $Q_{g,k}(t_i) \leftarrow$ Criticality of G_k at t_i from $Qlist$
27: **if** $Q_{g,k}(t_i) \ge \gamma_2$ **then**
28: $S_P(t_i) \leftarrow$ Hier_partitioning(G_k,t_i,$Q_{g,k}(t_i)$,γ_2)
29: $S_{PCP} \leftarrow S_{PCP} \cup S_P(t_i)$
30: **end if**
31: **end for**
32: **end if**
33: **end for**

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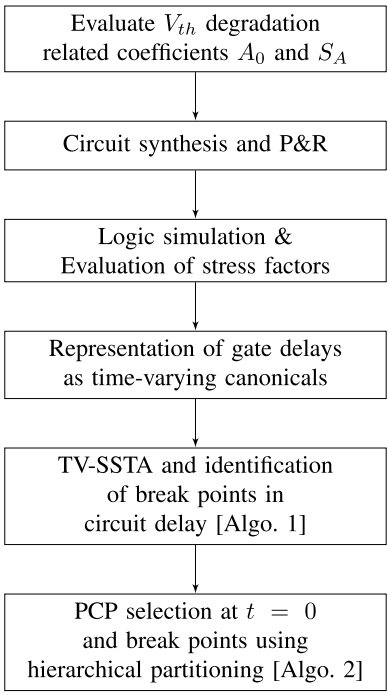

Fig. 3. Steps in the proposed framework.

union of critical paths identified at each of these partitioning steps.

VI. Proposed Framework

The various steps involved in the proposed framework is given in Fig. 3. The first step is to evaluate the threshold voltage degradation parameters A_0 and S_A as defined in Section II-A. These are constants for a given technology and operating conditions. Following this, circuit synthesis is performed to obtain the synthesized netlist. A place and route is also performed to obtain the location information of the gates. Given the synthesized netlist, a logic simulation is performed using 10^5 input vectors to generate the VCD file. The VCD file contains information regarding the signal transitions at the various nodes. This is processed to obtain the static signal probabilities and the correlation coefficients which are then used to evaluate the stress factors for the various pMOS transistors.

Once the stress factors have been computed, each of the edge delays is represented using the time-varying canonical delay model defined in (11). A time-varying forward SSTA is then performed to obtain the circuit delay distribution as a collection of time-varying canonicals with breakpoints in time. The MAX operation on time-varying canonicals is done as described in Algorithm 1. An extension of the hierarchical partitioning algorithm is then employed at $t = 0$ and at the breakpoints of the circuit delay distribution to identify the PCPs under aging. This is done as described in Algorithm 2.

VII. Results

The proposed algorithm for TV-SSTA and PCP selection is implemented in C++ and the experiments are performed on ISCAS'85, ISCAS'89, and ITC'99 benchmark circuits synthesized using United Microelectronics Corporation (UMC) 90-nm and NanGate 45-nm [27] technology libraries. The variations in three process parameters, L, W, and V_{th}, are considered along with NBTI for timing analysis. The standard deviation of each parameter variation is assumed to be 10% of the nominal value and 5% random variation is assumed. A three-level quad-tree is used for modeling the spatial correlations [28] and the variance is divided equally among the three layers. We aim to identify the PCPs with maximum global criticality above 0.05. From the experiments performed on various circuits to identify the critical paths at $t = 0$ using the hierarchical partitioning technique in [15], it was found that the average error in criticality is around 0.02 for both UMC 90-nm and NanGate 45-nm technologies. Therefore, the threshold Δ is fixed at 0.02. Accordingly, the criticality threshold is first set to 0.03 and is lowered to 0.01 for the second iteration. The results are verified by comparing against Monte Carlo simulations considering 10^5 samples. The circuit lifetime is assumed to be 10 years.

A. Time-Varying SSTA

Fig. 4 shows the percentage error in mean and standard deviation of circuit delay at the end of lifetime using the proposed TV-SSTA. The maximum error in mean of 3.17% occurs for s38417 circuit in UMC 90-nm technology and the average error is 0.70%. The maximum error in standard deviation is around 6% occurring in s9234 circuit synthesized using NanGate 4-nm library and the average error is 1.48%. If a conventional SSTA is performed using aged gate delays at $t = 10$ years, the error in mean is found to be 5.06% in s38417 circuit. Similarly, the error in standard deviation for s9234 circuit is 5.00%. This indicates that the error in these circuits is primarily due to the linearization of MAX operation in the standard SSTA formulation and there is no significant additional error due to the time-varying formulation.

Fig. 5 compares the time variation in mean and standard deviation of circuit delay against the values obtained by rerunning conventional SSTA multiple times, from 0 to 10 years in steps of 1 year for three circuits, c432, s13207, and b04, in UMC 90-nm technology. The mean of circuit delay is plotted as a function of time in Fig. 5(a). The figure indicates that the mean of circuit delay increases with time and matches closely with the values obtained from individual SSTA runs. The error in the mean of circuit delay at 10 years is found to be 0.02%, 0.07%, and 0.13% for c432, s13207, and b04 circuits, respectively. The time variation in standard deviation is plotted in Fig. 5(b). There is a discontinuity of 1.5% in standard deviation of s13207 which occurs at a break point in the circuit delay distribution. This is seen in some circuits and is typically less than 1%. The error in standard deviation of circuit delay at 10 years is found to be 0.62%, 0.04%, and 2.22% for c432, s13207, and b04 circuits, respectively.

The average number of breakpoints in arrival times across all the nodes ($N_{\text{tb,avg}}$) is shown in Fig. 6(a) and is found to be less than 2 in most of the benchmark circuits. This determines the run-time complexity of the TV-SSTA algorithm

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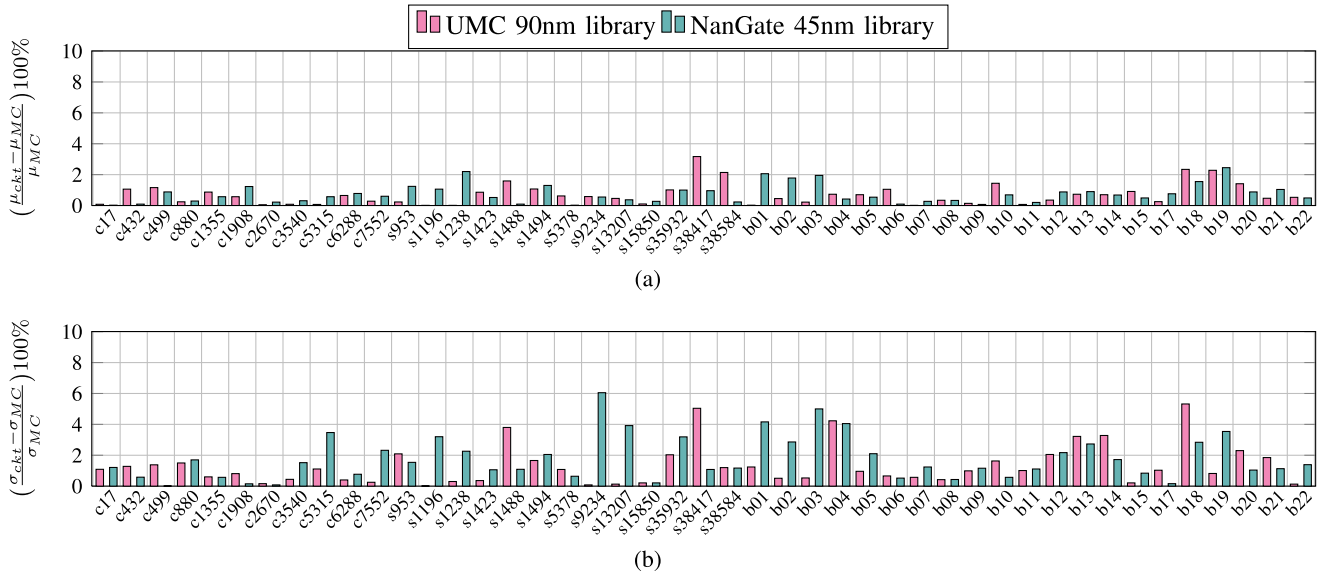

Fig. 4. Percentage error in mean and standard deviation of circuit delay at $t = 10$ years with respect to Monte Carlo simulations using the proposed TV-SSTA technique. (a) Percentage error in the mean of circuit delay. (b) Percentage error in standard deviation of circuit delay.

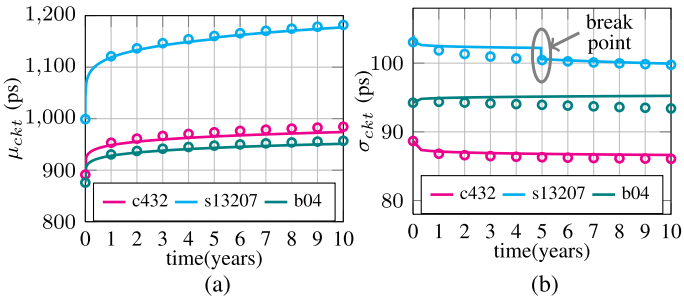

Fig. 5. Time variation in mean and standard deviation of circuit delay obtained using the proposed method in UMC 90-nm technology. "o" markers: values obtained using conventional SSTA runs. (a) Mean. (b) Standard deviation.

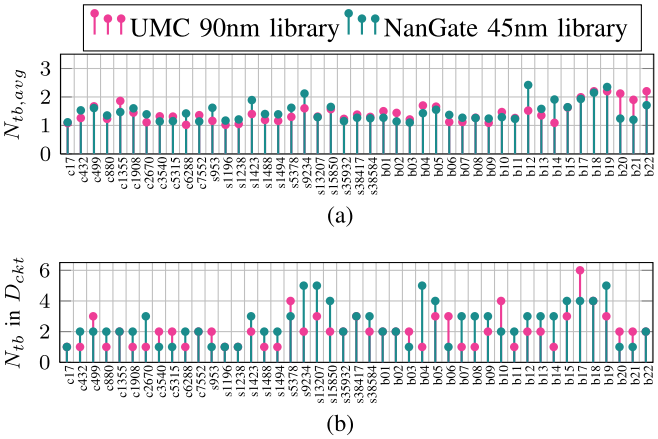

Fig. 6. Number of break points using TV-SSTA. (a) Average number of break points in arrival times across all nodes. (b) Number of break points in circuit delay.

and equals $O(N_{\text{tb,avg}}(|V| + |E|))$ which is $N_{\text{tb,avg}}$ times that of a conventional SSTA. On an average, the proposed SSTA algorithm is able to predict the time-varying circuit delay distribution in about $2\times$ the time required for a single SSTA run, with a maximum of around $3\times$ for larger circuits.

The number of breakpoints (N_{tb}) in circuit delay distribution is given in Fig. 6(b). The maximum number of breakpoints is six in the case of b17 circuit synthesized using UMC 90-nm technology and is less than four for most of the benchmark circuits. These breakpoints determine the time instants at which criticality evaluation and PCP selection will be done.

B. Time-Varying Path Criticality

To validate Property 5 in Section IV, we have considered a few circuits where there exists at least one path with non-monotonic criticality. Monte Carlo simulations are performed at the breakpoints in the circuit delay distribution and some test points and the criticality values are compared as shown in Fig. 7. The "x" markers in the figure denote the path criticalities obtained using Monte Carlo simulations at the breakpoints. The "o" markers denote the path criticalities obtained using Monte Carlo simulations at the test points $t \in \{0, 1, \ldots, 10\}$ years.

In s5378 circuit synthesized using UMC 90-nm technology, there are four break points in the circuit delay distribution at $t = 0.12, 1.45, 7.29$, and 10 years, as seen in Fig. 7(a). From the figure, we can see that the maximum criticality for the path occurs at some time instant between 0 and 1 year. Therefore, we have considered additional test points in this region, $t \in \{0.2, 0.4 \ldots, 0.8\}$ years and the results are shown in the inset figure. From the results, we can conclude that the criticality of path 1 is maximum at the breakpoint, $t = 0.12$ years. Similarly, in b05 circuit, there are three break points in the circuit delay distribution at $t = 0.02$, 0.1, and 10 years. The inset figure indicates that the maximum criticality for all the paths occur at the breakpoint $t = 0.1$ years. Similar results have also been obtained for circuits

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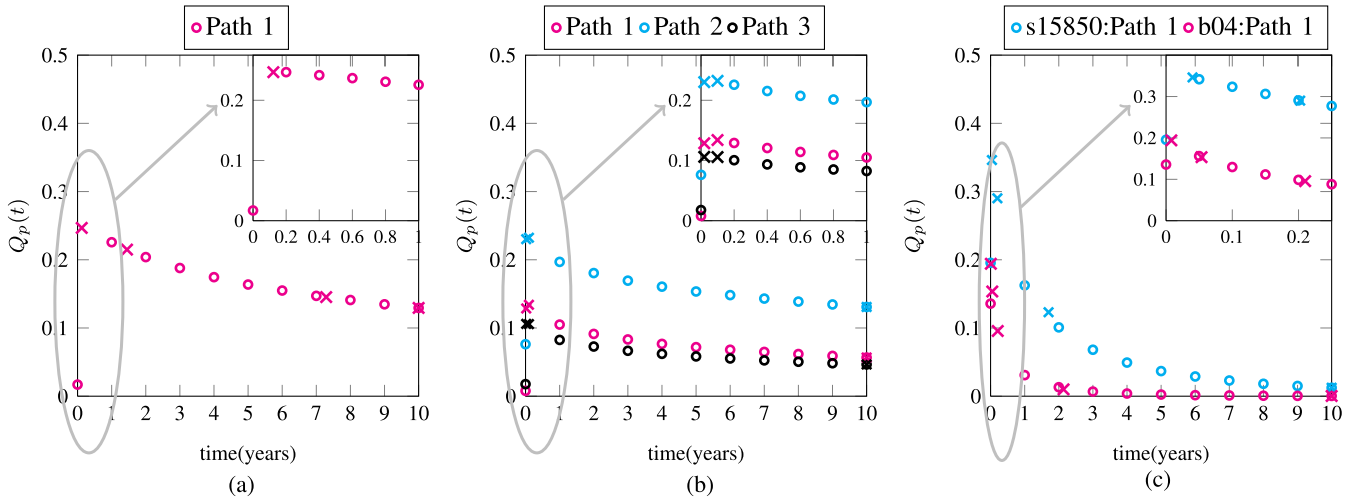

Fig. 7. Time-varying path criticality in s5378 (90 nm), b05 (90 nm), and s15850 and b04 (45 nm) circuits. "x" and "o" markers: Monte Carlo path criticalities at the break points in the circuit delay and the test points, respectively. A zoomed-in view of the region where maximum path criticality occurs is given in the inset figure. (a) s5378 circuit. (b) b05 circuit. (c) s15850 and b04 circuits.

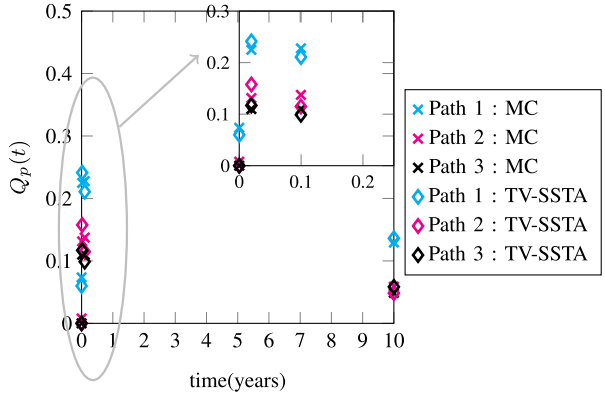

Fig. 8. Time-varying path criticality in b05 circuit synthesized using UMC 90-nm library. "x" and "◇" markers: path criticalities at the break points in circuit delay computed using Monte Carlo and hierarchical methods, respectively.

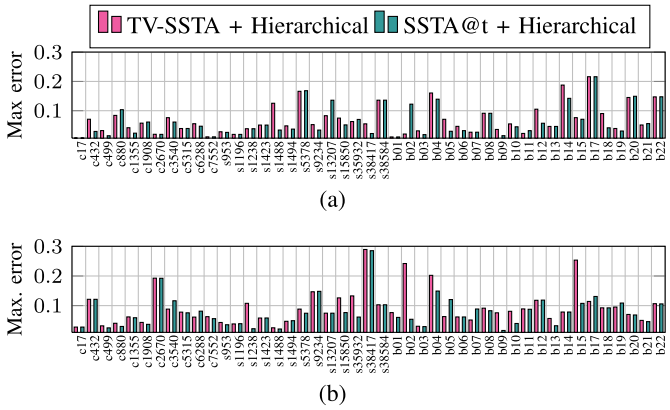

Fig. 9. Maximum error in path criticality with respect to Monte Carlo simulations for (a) TV-SSTA + hierarchical partitioning and (b) SSTA at a given time point + hierarchical partitioning in various benchmark circuits. (a) Using UMC 90-nm library. (b) Using NanGate 45-nm library.

synthesized using NanGate 45-nm library as seen in the case of s15850 and b04 circuits shown in Fig. 7(c).

It should be noted that the path criticality values plotted in Fig. 7 are obtained using Monte Carlo simulations. A comparison of the time variation of path criticality obtained using Monte Carlo simulations and TV-SSTA along with hierarchical partitioning is given in Fig. 8. It can be seen that, inspite of the errors involved, the overall trend is clear in most cases.

C. Statistical PCP Selection

Fig. 9 shows a comparison of the maximum error in path criticality computed using the proposed method and by performing a conventional SSTA followed by criticality computation at the break points. The aim is to see if any additional error is introduced due to the approximations made in the TV-SSTA formulation. The results show that the maximum error in both the cases is comparable, both for UMC 90-nm and NanGate 45-nm technologies, indicating that the approximations involved in the TV-SSTA formulation does not introduce any significant error in the path criticality. The maximum error in path criticality is between 0.2 and 0.3 and is found to be less than 0.1 in most of the benchmark circuits. Also, the average error in path criticality across all the break points equals 0.02 in both UMC 90-nm and NanGate 45-nm technologies.

The error in path criticality computations will affect the identification of PCPs. It should be noted that it is not just the maximum error, even a small error can push the criticality below the threshold. This is the reason why we have a guard band. Table I contains a list of circuits synthesized using UMC 90-nm technology in which the maximum error in path criticality is above 0.1. The time point at which the maximum error occurs along with the criticality values obtained using the proposed method (Q_{hier}) and Monte Carlo simulations (Q_{MC}) are also given in the table. The results indicate that, in most cases, the actual criticality values are well above the threshold so that the error does not cause the criticality to fall below the

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TABLE I

PATH CRITICALITY IN CIRCUITS WHERE THE MAXIMUM ERROR IS ABOVE 0.1 USING UMC 90-nm TECHNOLOGY

Ckt	Max error	Time	Q_{hier}	Q_{MC}
s1488	0.1255	10 yr	0.2552	0.3807
s5378	0.1663	10 yr	0.3899	0.5563
s38584	0.1360	0	0.2650	0.4010
b04	0.1605	10 yr	0.1784	0.0178
b14	0.1876	10 yr	0.0272	0.2148
b17	0.2161	0	0.2243	0.4403
b20	0.1452	4.11 yr	0.2934	0.4386
b22	0.1474	10 yr	0.2256	0.3730

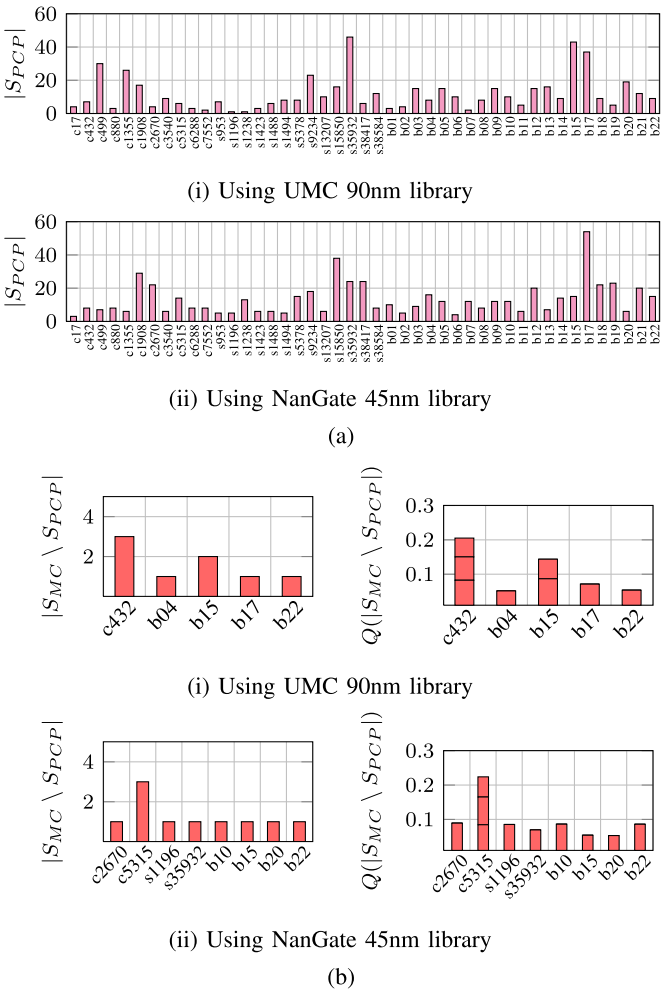

Fig. 10. Comparsion of paths reported by the proposed method against Monte Carlo simulations. (a) Number of PCPs reported by the proposed method. (i) Using UMC 90-nm library. (ii) Using NanGate 45-nm library. (b) Number of PCPs not identified by the proposed method with the associated maximum criticalities. (i) Using UMC 90-nm library. (ii) Using NanGate 45-nm library.

threshold. The only issue is in b14 circuit, and in this case as well, the path is identified when the threshold is lowered to 0.01.

To check the impact of error in path criticality on the identification of PCPs, we identified PCPs using both Monte Carlo and hierarchical methods and the comparison is shown in Fig. 10. In both the methods, the PCPs are identified by performing criticality computations at $t = 0$ and at the

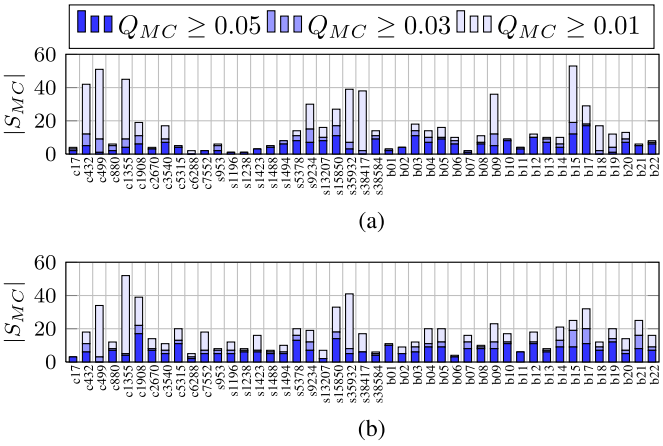

Fig. 11. Number of PCPs identified by Monte Carlo simulations for thresholds of 0.05, 0.03, and 0.01. (a) Using UMC 90-nm library. (b) Using NanGate 45-nm library.

breakpoints in the circuit delay distribution. Fig. 10(a) shows the number of PCPs obtained using the hierarchical method and Fig. 10(b) shows the number of PCPs with Monte Carlo criticality above 0.05 not identified by the proposed method in spite of lowering the threshold. It is a stacked plot indicating the maximum criticality of each of the missed paths. Note that the figure contains only those circuits in which the proposed method has failed to identify some PCPs. The results indicate that, in UMC 90-nm technology, a total of eight paths are missed by the proposed method with a maximum of three paths in c432 circuit. The maximum criticality missed is 0.08 in the case of b15 circuit. A similar trend has also been observed in the case of circuits synthesized using NanGate 45-nm library. Here, the number of paths missed by the proposed method is 10 with a maximum of three paths in c5315 circuit. The maximum criticality of the path missed is 0.09 in the case of c2670 circuit.

In order to check if the number of paths reported by our method is reasonable, we did a Monte Carlo analysis with three different thresholds (0.05, 0.03, and 0.01) and the results are given in Fig. 11. Note that in all the circuits, the number of paths identified using our method is greater than or equal to the number of paths reported by Monte Carlo analysis with the threshold set to 0.05. In our method, since we lower the threshold to compensate for the errors, we report a larger number of paths. This is especially seen in some circuits like c1355, s35932, and b15. This is because there are a large number of paths in these circuits with criticality in and around the threshold and we end up identifying all these paths.

VIII. CONCLUSION

In this paper, we have proposed a framework for TV-SSTA wherein the circuit delay distribution is obtained as a collection of time-varying canonicals valid over different periods with breakpoints in time. We proved that the global criticality of any path in the circuit will be maximum either at $t = 0$ or at the breakpoints in circuit delay distribution. This has also been verified using Monte Carlo simulations. As a result,

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the criticality computation and PCP selection need to be done only at these time points.

We have integrated the TV-SSTA along with a path selection and criticality computation technique proposed earlier to identify the PCPs under aging. Simulation results indicate that the proposed technique is able to identify the PCPs within a range of the specified threshold.

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